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Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 24-03-2026

PROJECT TITLE:REAL-TIME SYNC ENGINE

SDG: SDG 9 – Industry, Innovation and Infrastructure

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INTRODUCTION



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- In today's digital world, people need quick and easy communication without complex registration.
- Most existing chat applications require login, installation, or personal data sharing.
- This creates difficulty for temporary communication like classrooms, meetings, or quick discussions.
- There is a need for a simple system that allows instant communication with minimal effort.
- ChatMeet solves this problem by enabling users to create and join chat rooms using a 4-digit code.
- It eliminates the need for sign-up, making it fast, accessible, and user-friendly.
- The system uses real-time communication to ensure instant message delivery between users.
- Overall, ChatMeet provides a lightweight, efficient, and practical solution for temporary group communication.

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PROBLEM STATEMENT



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➤ In many situations, people need instant communication without creating accounts or installing applications.

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➤ Existing chat platforms require registration, personal details, and complex setup processes.

➤ This makes it difficult for temporary communication such as classrooms, group discussions, or quick meetings.

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➤ Users also face privacy concerns when sharing personal information unnecessarily.

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➤ There is no simple solution for creating a temporary chat space quickly and easily.

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➤ This results in delays, inconvenience, and reduced efficiency in communication.

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➤ Therefore, there is a need for a lightweight and user-friendly system for instant communication.

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➤ The system should allow users to join and communicate easily without registration.

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OBJECTIVE(S)



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1: System Design and Development

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To design and develop a user-friendly real-time chat application that enables users to create

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and join chat rooms using a unique 4-digit code, eliminating the need for registration and ensuring quick and easy access for temporary communication.

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2: Implementation of Real-Time Features

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To implement and integrate advanced real-time communication features such as instant messaging, file and image sharing, emoji support, typing indicators, and admin controls (mute/kick), ensuring efficient and interactive user communication.

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PROPOSED SYSTEM



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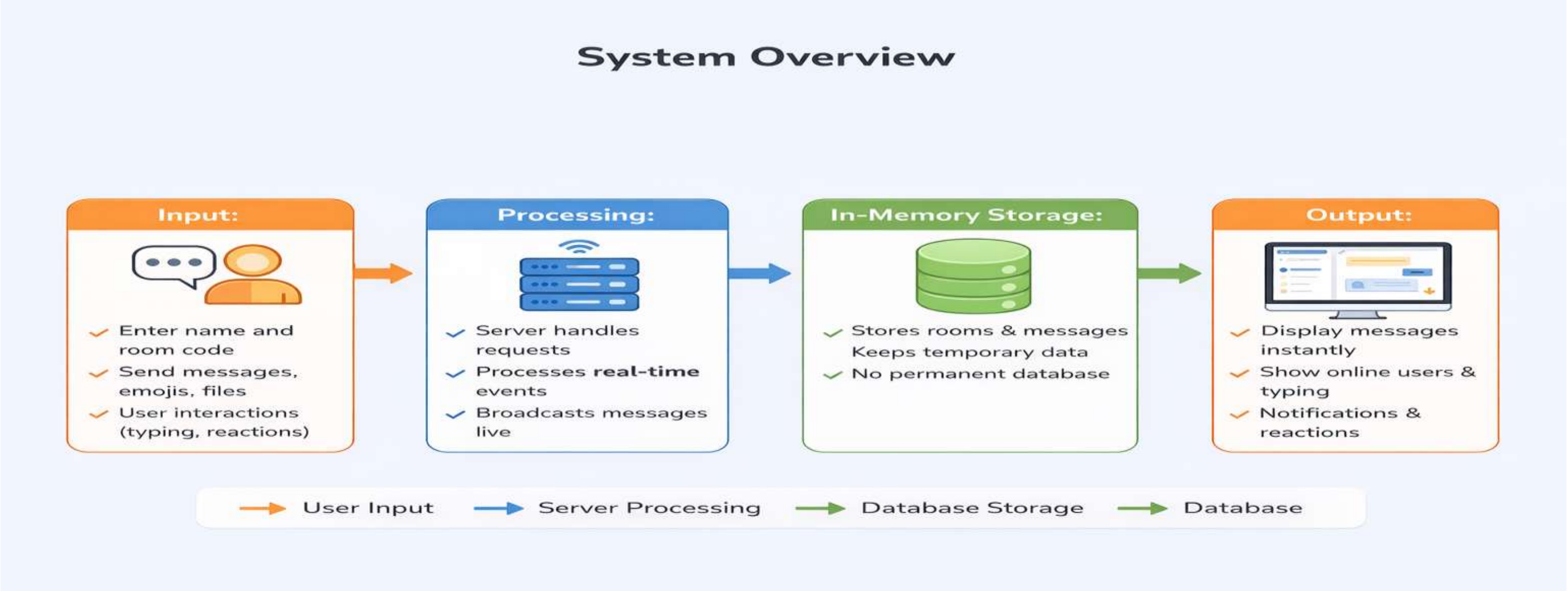
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- ChatMeet provides a web-based platform where users can instantly create or join chat rooms using a unique 4-digit code.
- It removes the need for registration, login, or sharing personal details, making access quick and simple.
- The system uses real-time communication (Socket.IO) to ensure instant message delivery between users.
- Users can share text, emojis, images, and files, making communication more interactive and effective.
- A typing indicator improves user experience by showing real-time activity during conversations.
- Host controls such as mute and kick allow better management of users within the chat room.
- Compared to existing systems, ChatMeet is lightweight, faster, and does not require installation.
- Overall, it offers a convenient, secure, and efficient solution for temporary group communication.

PROPOSED SYSTEM ARCHITECTURE



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User Input → Server Processing → In-Memory Storage → Real-Time Output

TECHNOLOGIES USED



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Frontend:

HTML5 , CSS3 , Bootstrap , JavaScript

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Backend:

Node.js , Express.js , Socket.IO

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Backend:

Node.js , Express.js , Socket.IO

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Database:

- In-Memory Storage (JavaScript Objects)
- *(No permanent database used)*

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Tools:

- Visual Studio Code
- Git & GitHub
- Render (Backend Hosting)
- GitHub Pages (Frontend Hosting)

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Hardware/Software Requirements:

Hardware:

- Minimum 4GB RAM
- Any modern computer or laptop

Software:

- Web Browser (Chrome, Edge, Firefox)
- Node.js installed
- Internet connection

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MODULE DESCRIPTION



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Module 1: Room & User Management

Handles room creation and joining using a unique 4-digit code.

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Manages user names, participant list, and host identification without requiring registration.

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Module 2: Real-Time Communication

Implements real-time messaging using Socket.IO.

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Processes user inputs like text messages, emojis, file sharing, and typing indicators efficiently.

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Module 3: Data Management (In-Memory)

Stores chat messages, user details, and room information temporarily using in-memory storage.

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Ensures quick access and updates during the active session.

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Module 4: Admin Control & Interaction

Provides host controls such as mute and kick for managing users.

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Handles user interactions like notifications, typing status, and message updates.

IMPLEMENTATION SCREENSHOTS



Home Page (index.html)

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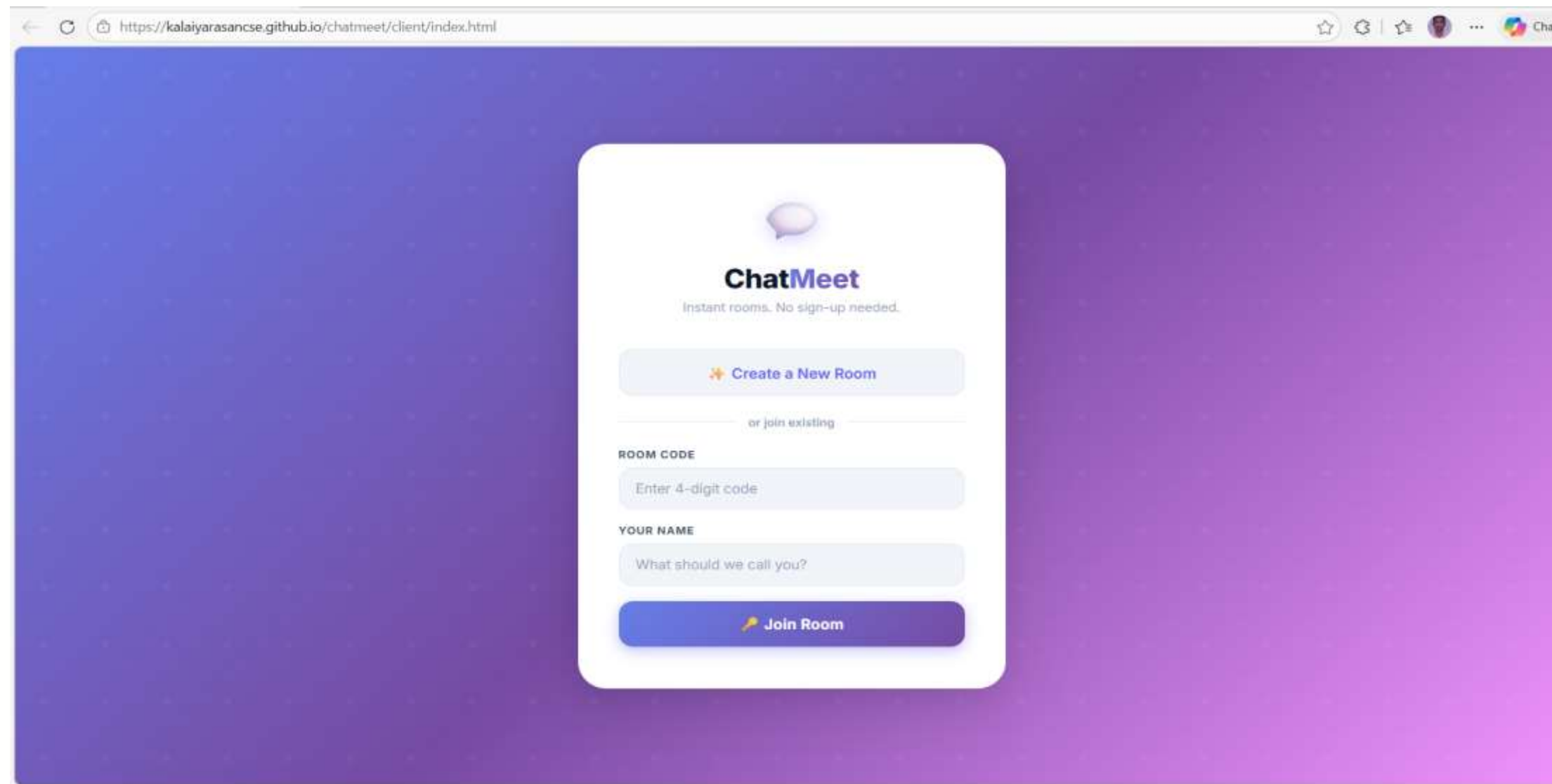
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Allows users to create a new chat room or join an existing room using a 4-digit code .It collects the user name and directs them to the appropriate chat room.

IMPLEMENTATION SCREENSHOTS



Create Room Page (create.html)

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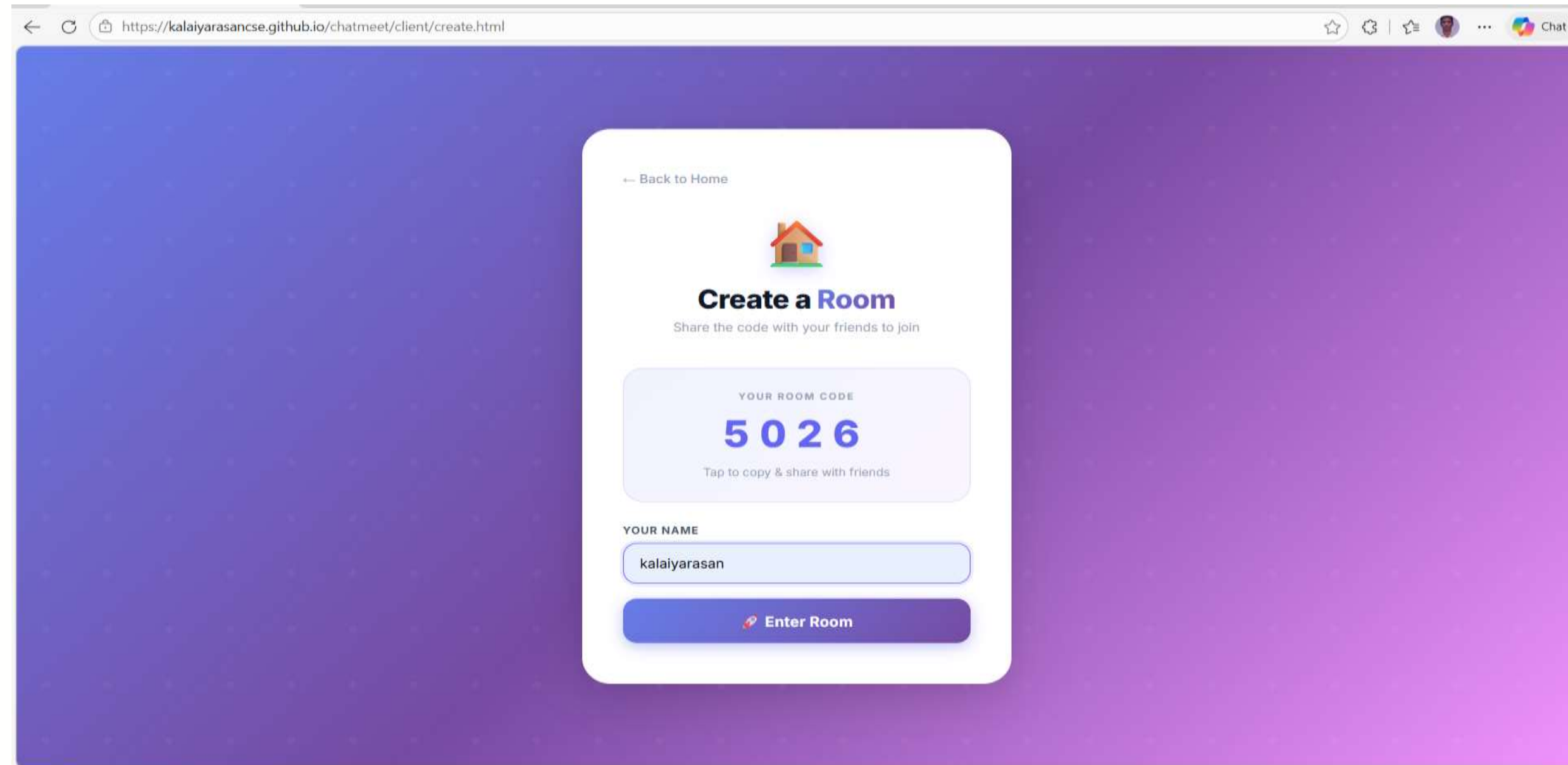
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Enables the user to create a new chat room by generating a unique 4-digit code. The generated code can be shared with others to join the chat session.

IMPLEMENTATION SCREENSHOTS



Chat Room Page (room.html)

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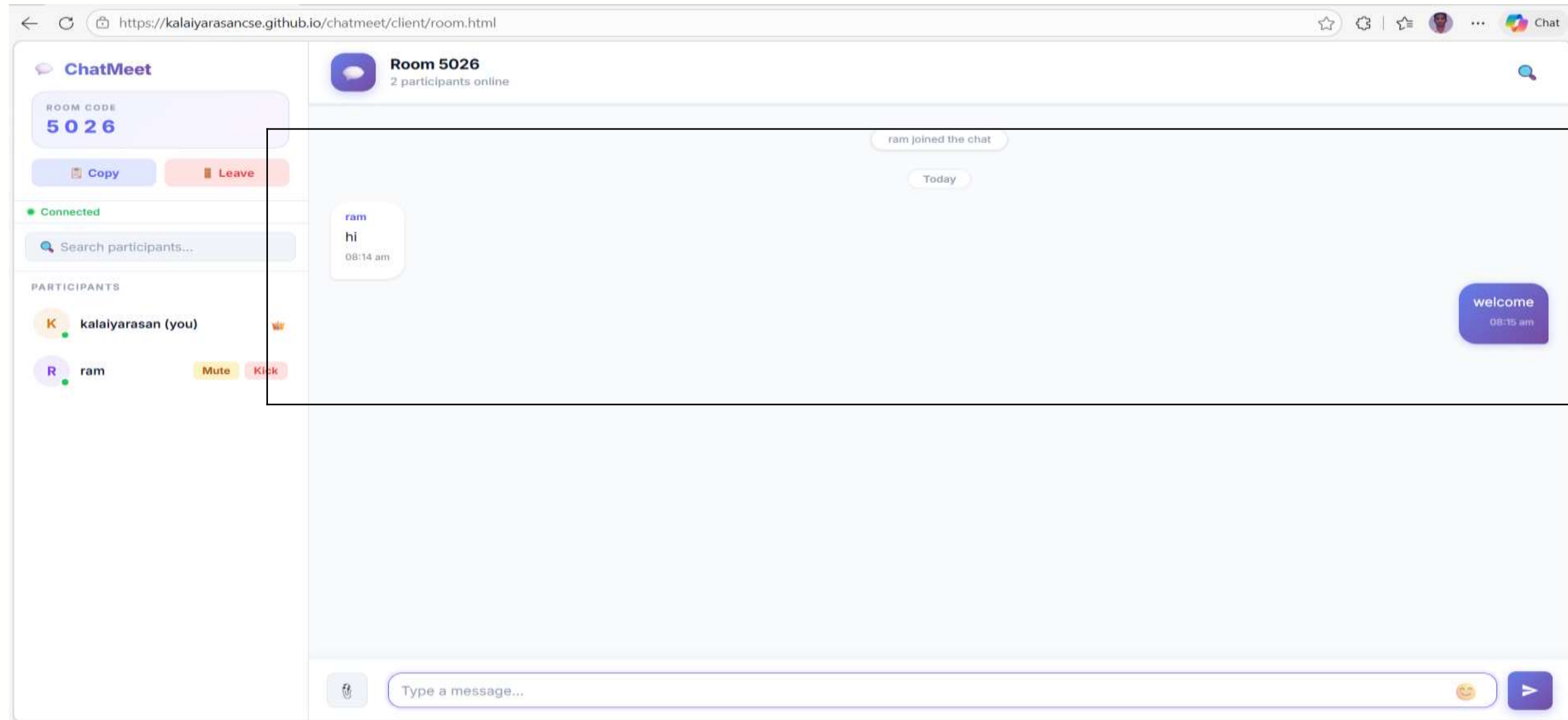
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Provides real-time communication features such as messaging, emoji, and file sharing. Displays participants and allows the host to manage users with controls like mute and kick.

IMPLEMENTATION SCREENSHOTS



User Login with Same ID

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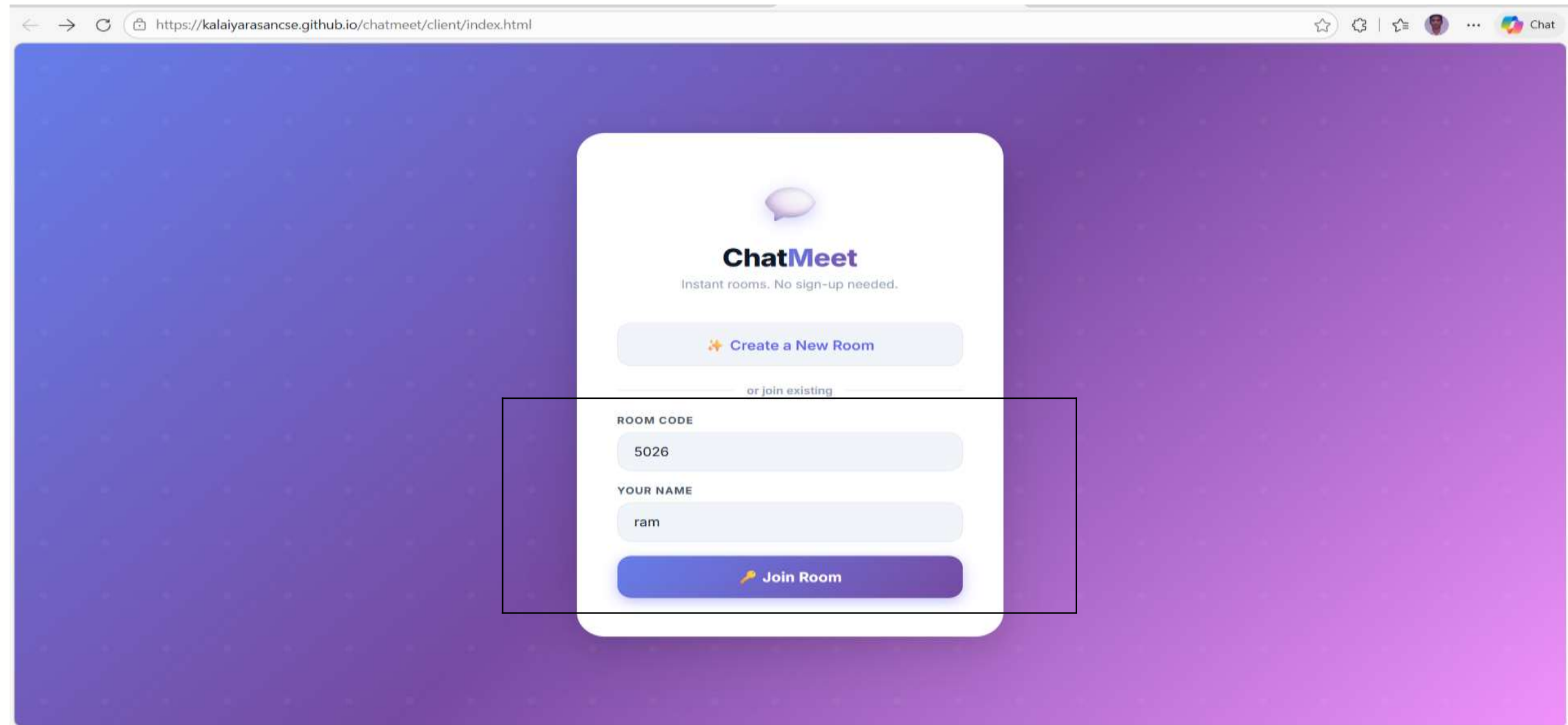
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Multiple users can join the same chat room using a shared 4-digit room code. Each user is identified by their chosen name within the room, even if they use the same code.

ADVANTAGES & LIMITATIONS



Advantages

- No registration required, making it quick and easy to use
- Instant room creation using a simple 4-digit code
- Real-time communication using Socket.IO
- Supports file sharing, emojis, and typing indicators
- Works directly in a web browser (no installation needed)
- Lightweight and fast performance
- Host controls for managing users (mute/kick)
- Free hosting and easy deployment

Limitations

- Data is stored in-memory and lost after server restart
- No permanent user accounts or chat history
- Limited scalability for very large number of users
- Requires internet connection to function
- No advanced security features (like encryption/login)
- File sharing size may be limited
- No voice or video calling support

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Thank you

